P R O V E

Cal Poly San Luis Obispo — Mechanical Team

PROJECT OUTLINE

Project Documentation Template

Project Name:	Electronics Control Panel
Subsystem:	Mechanical, Electrical
Team Members:	Kaylee Wong, Hannah Redoblado
Vehicle:	MILA
Document Version:	2.0
Date Started:	3/3/26
Status:	*In Progress*

This document serves as the official post-project record for the above project. It captures goals, requirements, design decisions, manufacturing, testing results, and lessons learned for future reference.

1 Project Overview

1.1 Project Summary

The Modular Electronics Control Panel is a centralized mounting system designed to house high-voltage (HV) and low-voltage (LV) control electronics for the university's electric sports car. It solves the issue of permanent, inflexible component mounting, allowing for rapid maintenance, accessibility, and reconfiguration without mechanical redesign. This modularity is critical for minimizing vehicle downtime during testing while ensuring organized, safe cable routing.

1.2 Objectives & Goals

- Create a "double-layered" mounting interface that allows for the removal of the entire base panel from the car, as well as the independent removal and positional adjustment of individual components without needing to drill new mounting holes
- Consolidate all components within a 19.5" x 3.5" footprint using non-conductive, UL94-V0 rated materials (like FR4 or isolated Polycarbonate) to prevent electrical shorts
- Establish a layout that prioritizes organized cable paths, fanning out connections to minimize wire crossover and simplify troubleshooting
- Ensure the mounting system remains stable and secure under the high-vibration and high-G environments typical of car dynamics
- Target Deadline: Complete by the end of Spring Quarter 2026

1.3 Scope

This project serves as the primary interface between the vehicle's electrical logic and physical chassis. It integrates directly with the HV Battery Box (side-mounted) and manages the layout for at least four major PCB subsystems (Logic, Relay, Charger, and Power Distribution). It interacts with the Inverter, Motor, and Dashboard via front-to-back wiring harnesses and connects the LV battery input to the various control boards.

1.4 Background & Motivation

The primary motivation is to develop a modular housing solution that empowers the HV and LV teams to modify their configurations as needed with minimal mechanical intervention. By creating a flexible mounting standard, the team can iterate electrical designs and swap components on the fly, ensuring the car remains adaptable throughout the design and testing phases.

2 Requirements

2.1 Structural Constraints

- Dimensions: The entire assembly must not exceed 19.5” (W) x 14.375” (H) x 3.5” (D).
- Clearance: A minimum of 6mm standoff is required between the PCB surface and the carrier/baseplate to allow for rear-mounted component clearance and passive airflow.
- Mounting Interface: The base panel must mount to the side of the HV battery box using existing chassis mounting points, with tolerances capable of being met by manual alignment during installation.

2.2 Subsystem Interface Requirements

- The base panel must be constructed from non-conductive materials to provide a secondary insulation barrier between HV components and the vehicle chassis.
- The system must allow the HV and LV teams to relocate any of the 10+ electrical components without drilling new holes in the base panel or chassis.
- The entire control panel must be removable as a single unit for off-car troubleshooting, while individual carriers must be independently removable for bench testing.

2.3 Safety Requirements

- All mounting materials must be UL94-V0 rated.
- The Manual Service Disconnect (MSD) must remain the most accessible component, requiring zero tools for operation or emergency removal??
- All mechanical fasteners must utilize secondary locking features (e.g., Nyloc nuts or Loctite 243) to prevent loosening due to high-frequency vehicle vibrations.

2.4 Performance Requirements

- The mounting system must maintain component position under 3G lateral and longitudinal loads without sliding or deflection.
- The layout must have fanned-out connections from the Power Distribution PCB (PDC) to minimize wire crossover, reducing heat concentration and electromagnetic interference (EMI).
- As the components are "light," the total weight of the mounting infrastructure (carriers and rails) should be minimized to avoid shifting the vehicle’s center of gravity.

2.5 Requirements Summary Table

Req. ID	Category	Requirement	Source	Verification Method	Met?
REQ-001	**Structural**	Withstand weight of all components	FEA	Analysis	✓
REQ-002	**Performance**	All components securely mounted, passes driving test	Drive Test	Inspection	

REQ-003	**Subsystem Interface**	Design and prototype approved by HV and LV	HV and LV Feedback	Feedback	
REQ-004	**Safety**	Organized wire routing and vibration resistant	Subsystem Y Interface Doc	Inspection	
REQ-005	**Safety**		PROVE Safety Guidelines	Inspection	
REQ-006	**Performance**		Vehicle Targets Doc	Test	
REQ-007	**Performance**		Vehicle Targets Doc	Test	

Verification Methods: Analysis | Inspection | Test | Demonstration
Categories: Structural | Subsystem Interface | Safety | Performance

3 Design

3.1 Design Alternatives Considered

Describe the different design options that were evaluated before arriving at the final design. Use the table to compare alternatives.

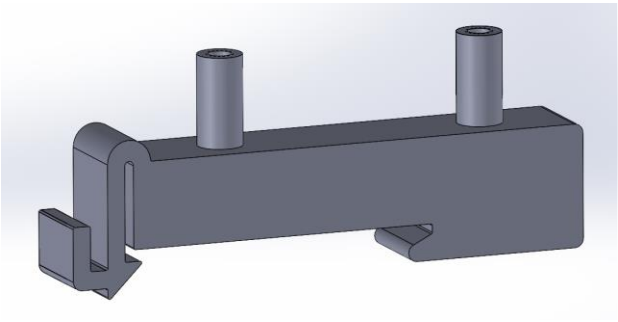
Control Panel Mounting System

Option	Description	Pros	Cons	Selected?
A	DIN Rails with 3D printed Mounts	Easily reusable	Bit more expensive	Yes
B	Peg hole panel with 3mm threaded holes	Easy to manufacture	Need perfect tolerances, easy to strip threads after constant use	No

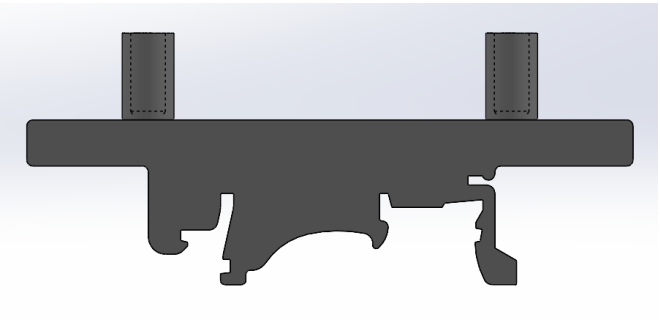
Electronic Component Mounts

Option	Description	Pros	Cons	Selected?
A	Spring DIN Rail mounts, clamp on	Commercial part so easy to purchase multiple	Bit more expensive, will spring wear with overuse?	No
B	3D Printed DIN Rail Mounts, slides on	Easily customizable	Not as study material	Yes

Mount Design Iteration 1:



Mount Design Iteration 2 (Selected):

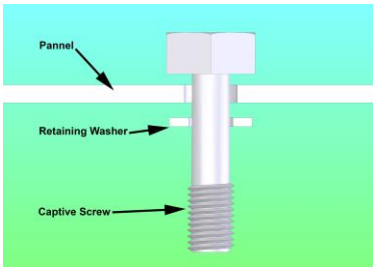


Cable Routing

Option	Description	Pros	Cons	Selected?
A	Screw-on Cable Saddle	Commercial part	Requires another mount?	TBD

				
B	DIN Rail cable clamp 	Easy to customize, holds multiple cables in one assembly	Customized part (requires more time)	Yes

Mounting Polycarbonate Panel to Battery Box

Option	Description	Pros	Cons	Selected?
A	Guide Rails 	Secure, smooth sliding	$$$, thick	No
B	C Rails 	Ball bearings! Smooth sliding, thinner than guide rails	$$$	No
	M5 Captive Screws with Rivnuts into MDF 	Easy to assemble, commercial parts	More prone to human error	Yes

3.2 Design Selection & Rationale

A combination of sliding and clamping mechanisms for PCBs and screws for heavier components to attach HV and LV components allows for the simplest and most user-friendly mounting system. The PCBs can simply be slid on and off and reorganized with standardized spacers and stoppers. This

mechanism allows the HV and LV teams to easily adjust the configuration of the control panel as needed by simply

3.3 Final Design Description

Provide a complete description of the final design. Describe geometry, materials, key dimensions, and how the component functions. Include images, CAD screenshots, and drawings below.

▶ 3.3.1 CAD Model & Drawings

[Insert CAD model screenshots here (front, side, isometric views). Reference part numbers and drawing numbers.]

▶ 3.3.2 Key Dimensions & Parameters

- *Key dimensions: Control Panel 470mm tall, 520mm wide,*
- *Material: Polycarbonate Panel, PLA 3D Printed Mounts, Stoppers, and Spacers*
- *[Mass: (e.g., 0.42 kg)]*
- *[Factor of Safety: (e.g., 2.1 @ max load)]*

3.4 Analysis & Simulation

Document any calculations, FEA, CFD, or other analyses performed to validate the design. Attach or reference detailed analysis reports in the appendix.

▶ 3.4.1 Hand Calculations

[Summarize key hand calculations (e.g., buckling load, stress at critical section). Reference full calculations in Appendix A.]

▶ 3.4.2 FEA / Simulation Results

[Insert FEA results here — mesh, boundary conditions, max stress, deformation, factor of safety. Reference Appendix B for full reports.]

4 Bill of Materials (BOM)

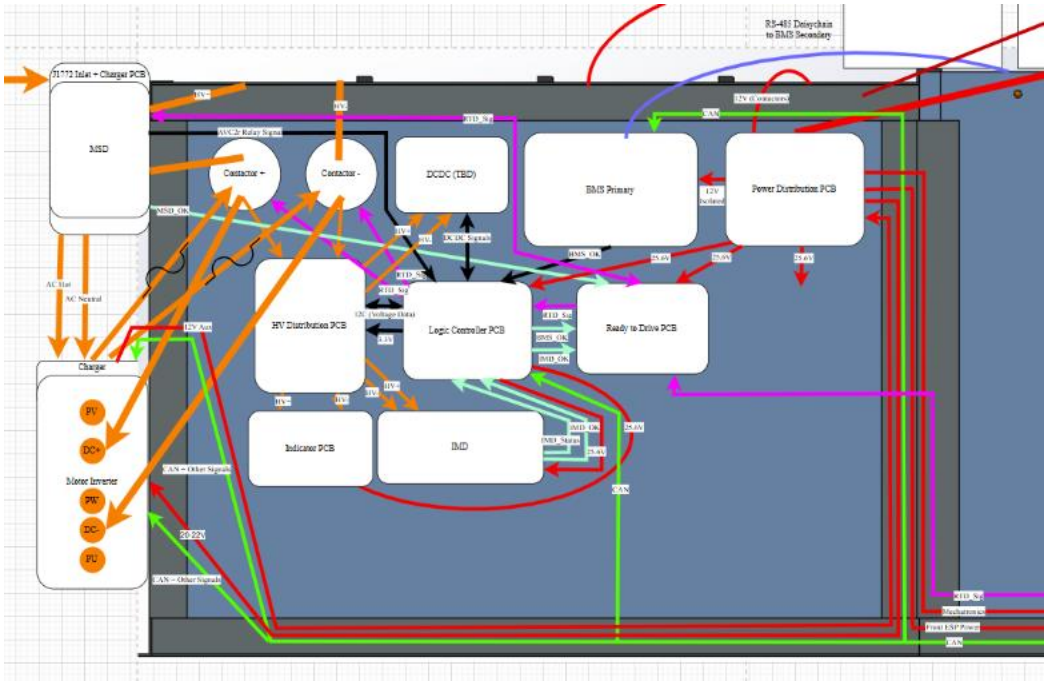
List all purchased and manufactured parts for this project. Include supplier, cost, and lead times.

Item	Part Name & Link	Part #	Supplier	Qty	Unit Cost	Total Cost	Lead Time
1	Contactor	-	HV	2	-	-	-
2	BMS Primary	-	HV	1	-	-	-
3	IMD	-	HV	1	-	-	-
4	Indicator PCB	-	HV	1	-	-	-
5	HV Distribution PCB	-	HV	1	-	-	-
6	Logic Controller PCB	-	HV	1	-	-	-
7	Power Distribution PCB	-	HV	1	-	-	-
8	Ready to Drive PCB	-	HV	1	-	-	-
9	Motor Invertor	-	HV	1	-	-	-
10	MSD	-	HV	1	-	-	-
11	Polycarbonate Panel	410NZ4	Grainger	1	$95.54	$95.54	2 weeks
12	35 mm Din Rails	8961K94	McMaster	3	$6.07	$18.21	1 week
14	3D Printed PCB Mount		Makerspace	5	$0	$0	3 days
15	3D Printed Contactor Mounts		Makerspace	2	$0	$0	3 days
16	3D Printed BMS Mount		Makerspace	2	$0	$0	3 days
17	3D Printed IMD Mount		Makerspace	1	$0	$0	3 days
18	DIN Rail Stoppers	5575N21	McMaster	6 (1 pk of 10)	$11.43	$11.43	3 days
19	DIN Rail Spacers		Makerspace	TBD	$0	$0	3 days
20	M4 brass heat-set inserts Kit		Amazon	1	$15.99	$15.99	2 days
21	M4 x 8mm Screws (4 bolts per rail = 12 max)		Garage	12	$8.91	$0	
22	M4 Shoulder Washer (DIN Rails)		Garage	12	$13.10	$0	
23	14 mm M4 Aluminum Standoffs (for heavier components)	98952A041	McMaster	8	$0.95	$7.60	
24	M5 stainless rivnuts for captive screw to mount panel	B0780520	Mustang 60?	6 (1 pack of 10)	$10.93	$10.93	
25	M5 Captive Screw Assembly		Misumi (need 6)	1 pk of 11	$15.49	$15.49	
26	M4 Cable saddle (for HV wires)	Later					

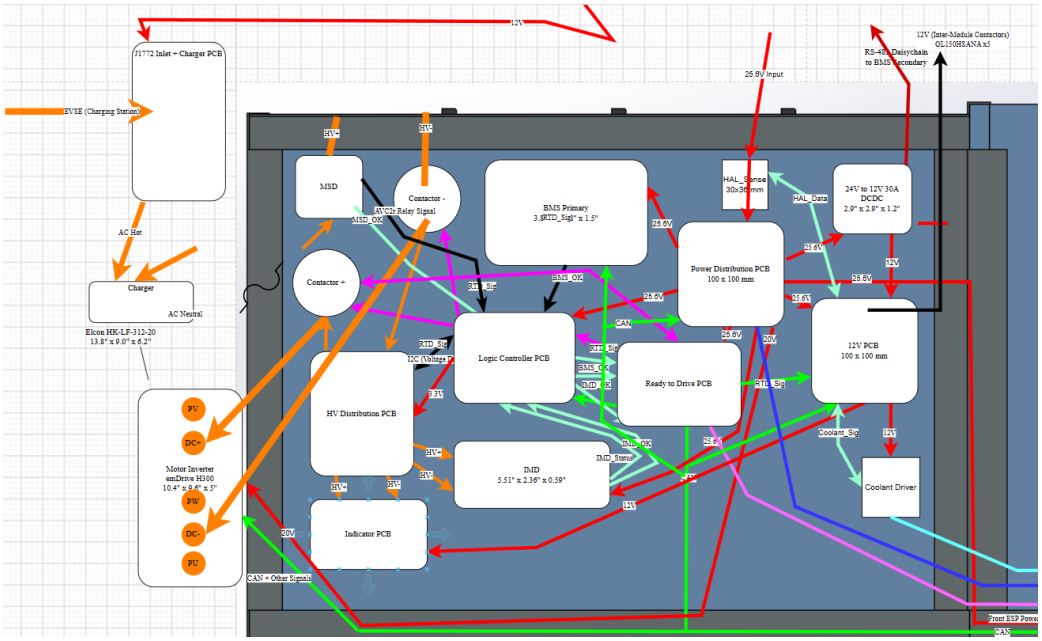
27	Mini Zip Ties	GT-100MB			$7.61	$7.61	

Total Estimated Cost: *$200* **Actual Cost:** *$167.18*

Old Diagram



Updated Diagram



5 Manufacturing & Assembly

5.1 Manufacturing Process Details

Summarize the manufacturing processes used in this project (e.g., CNC milling, waterjet, welding, carbon fiber layup, 3D printing).

▶ 5.1.1 [Polycarbonate Panel]

The Table saw and a standard power drill will be used to cut the panel to length and drill holes. All components will be mounted to the panel via the DIN rail or screws into brass heat-set inserts. Heavier components, such as the BMS, will be mounted using M4 Aluminum Standoffs.

- Mounting DIN Rails to Polycarb: M4 heat-set inserts, 4 screws per rail with shoulder washers
- Mounting Heavier components: M4 Aluminum Standoffs into brass heat-set inserts
- Mounting Polycarbonate panel to MDF Battery Box: M5 Captive Screws and Rivnuts

▶ 5.1.2 [3D Printed Mounts]

All mounts will be 3D printed out of PETG, which has a good balance of rigidity, toughness, and heat resistance. The mounts, spacers, and hard stops will be printed on the side to capture all the important features and improve structural rigidity. This orientation also allows for layer lines to run parallel to the primary load direction.

▶ 5.1.3 [Threaded Inserts]

To screw on PCBs onto their respective mounts, threated inserts will be soldered onto the top of the mount.

5.2 Manufacturing Challenges & Solutions

[Describe any significant manufacturing challenges encountered and how they were resolved. This is valuable institutional knowledge for future teams.]

[Insert photos of finished assembly]

6 Testing & Validation

6.1 Test Plan Overview

Summarize the test plan. What tests were conducted to verify the design meets its requirements?

[Overview of testing approach — bench tests, vehicle tests, static load tests, etc.]

6.2 Test Results

Test ID	Test Name	Requirement	Pass Criteria	Result	Pass/Fail	Notes
T-001		REQ-001			**Pass**	
T-002		REQ-002			**Fail**	

6.3 Test Details & Data

▶ 6.3.1 [Test Name]

[Describe the test setup, procedure, instrumentation, and results. Insert data plots, photos, or graphs.]

▶ 6.3.2 [Test Name]

[Repeat for additional tests]

6.4 Validation Summary

State clearly whether the design met all requirements. If any requirements were not met, describe the corrective action taken or the risk accepted.

[Final validation statement — did the design pass all requirements? Note any open items.]

7 Lessons Learned & Recommendations

7.1 Lessons Learned

Be honest and thorough here — this section is the most valuable for future teams. Capture what you wish you had known at the start.

Category	What Worked	What Didn't Work	Recommendation for Next Year
Design			
Manufacturing			
Testing			
Teamwork / Process			

7.2 Future Recommendations

What would you recommend the next team working on this system do differently? Are there known weaknesses or areas for improvement?

- *[Recommendation 1]*
- *[Recommendation 2]*
- *[Known issues or open design items for future work]*

8 References

List all references used in this project: textbooks, papers, supplier datasheets, previous PROVE documentation, SAE papers, etc.

1. *[Author], [Title], [Source], [Year]*
2. *[Supplier Datasheet] — [Part Name], retrieved [date]*

Appendices

Appendix A — Detailed Calculations

[Attach or embed detailed hand calculations, derivations, or analytical models here]

Appendix B — FEA / Simulation Reports

[Full FEA reports, mesh details, load cases, and result plots]

Appendix C — Engineering Drawings

[Insert dimensioned engineering drawings for all manufactured parts (PDF or embedded image)]

Appendix D — Photos & Media

[Additional photos of design, manufacturing process, assembly, and testing not included in the main body]

Appendix E — Revision History

Version	Date	Author	Description of Changes
1.0			Initial document